

```
In [2]: import numpy as np
import pandas as pd
%matplotlib inline
import matplotlib.pyplot as plt
```

```
In [3]: avocado = pd.read_csv(r'D:\AI ML\FSDS AI\FSDS AI CLASS DOCS AND CODE\Dec\26th - dtr
```

```
In [5]: avocado.head()
```

```
Out[5]:
```

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26

```
In [6]: avocado = avocado.reset_index(drop=True)
avocado.head()
```

```
Out[6]:
```

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26

```
In [9]: avocado = avocado.reset_index(drop=True)
avocado.head()
```

Out[9]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26

In [10]: `avocado.to_csv("test_write.csv")`In [12]: `avocado = pd.read_csv(r"D:\AI ML\FSDS AI\FSDS AI CLASS DOCS AND CODE\Dec\26th - dtr
avocado.head()`

Out[12]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26

In [13]: `avocado.tail(10)`

Out[13]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Sr B
18239	2	2018-03-11	1.56	22128.42	2162.67	3194.25	8.93	16762.57	16510
18240	3	2018-03-04	1.54	17393.30	1832.24	1905.57	0.00	13655.49	13400
18241	4	2018-02-25	1.57	18421.24	1974.26	2482.65	0.00	13964.33	13690
18242	5	2018-02-18	1.56	17597.12	1892.05	1928.36	0.00	13776.71	13550
18243	6	2018-02-11	1.57	15986.17	1924.28	1368.32	0.00	12693.57	12430
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13060
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9350
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10910
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11980

In [14]: avocado.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18249 entries, 0 to 18248
Data columns (total 14 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      18249 non-null  int64
1   Date            18249 non-null  object
2   AveragePrice    18249 non-null  float64
3   Total Volume    18249 non-null  float64
4   4046            18249 non-null  float64
5   4225            18249 non-null  float64
6   4770            18249 non-null  float64
7   Total Bags      18249 non-null  float64
8   Small Bags      18249 non-null  float64
9   Large Bags      18249 non-null  float64
10  XLarge Bags     18249 non-null  float64
11  type            18249 non-null  object
12  year            18249 non-null  int64
13  region          18249 non-null  object
dtypes: float64(9), int64(2), object(3)
memory usage: 1.9+ MB

```

In [15]: `print(avocado.shape)`

(18249, 14)

In [16]: `avocado.describe()`

Out[16]:

	Unnamed: 0	AveragePrice	Total Volume	4046	4225	477
count	18249.000000	18249.000000	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+04
mean	24.232232	1.405978	8.506440e+05	2.930084e+05	2.951546e+05	2.283974e+05
std	15.481045	0.402677	3.453545e+06	1.264989e+06	1.204120e+06	1.074641e+06
min	0.000000	0.440000	8.456000e+01	0.000000e+00	0.000000e+00	0.000000e+00
25%	10.000000	1.100000	1.083858e+04	8.540700e+02	3.008780e+03	0.000000e+00
50%	24.000000	1.370000	1.073768e+05	8.645300e+03	2.906102e+04	1.849900e+04
75%	38.000000	1.660000	4.329623e+05	1.110202e+05	1.502069e+05	6.243420e+04
max	52.000000	3.250000	6.250565e+07	2.274362e+07	2.047057e+07	2.546439e+07



In [17]: `avocado.values`

Out[17]: array([[0, '2015-12-27', 1.33, ..., 'conventional', 2015, 'Albany'],
 [1, '2015-12-20', 1.35, ..., 'conventional', 2015, 'Albany'],
 [2, '2015-12-13', 0.93, ..., 'conventional', 2015, 'Albany'],
 ...,
 [9, '2018-01-21', 1.87, ..., 'organic', 2018, 'WestTexNewMexico'],
 [10, '2018-01-14', 1.93, ..., 'organic', 2018, 'WestTexNewMexico'],
 [11, '2018-01-07', 1.62, ..., 'organic', 2018, 'WestTexNewMexico']],
 dtype=object)

In [18]: `print(avocado.columns)`

Index(['Unnamed: 0', 'Date', 'AveragePrice', 'Total Volume', '4046', '4225',
 '4770', 'Total Bags', 'Small Bags', 'Large Bags', 'XLarge Bags', 'type',
 'year', 'region'],
 dtype='object')

In [22]: `avocado.sort_values(["AveragePrice", "year"], ascending=[True, False])`

Out[22]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	
15261	43	2017-03-05	0.44	64057.04	223.84	4748.88	0.00	5908
7412	47	2017-02-05	0.46	2200550.27	1200632.86	531226.65	18324.93	45036
15473	43	2017-03-05	0.48	50890.73	717.57	4138.84	0.00	4605
15262	44	2017-02-26	0.49	44024.03	252.79	4472.68	0.00	3929
1716	0	2015-12-27	0.49	1137707.43	738314.80	286858.37	11642.46	10089
...	...	...	...	...	...	...	...	...
16720	18	2017-08-27	3.04	12656.32	419.06	4851.90	145.09	724
16055	42	2017-03-12	3.05	2068.26	1043.83	77.36	0.00	94
14124	7	2016-11-06	3.12	19043.80	5898.49	10039.34	0.00	310
17428	37	2017-04-16	3.17	3018.56	1255.55	82.31	0.00	168
14125	8	2016-10-30	3.25	16700.94	2325.93	11142.85	0.00	325

18249 rows × 14 columns



In [24]: avocado["AveragePrice"]

Out[24]:

0	1.33
1	1.35
2	0.93
3	1.08
4	1.28
	...
18244	1.63
18245	1.71
18246	1.87
18247	1.93
18248	1.62

Name: AveragePrice, Length: 18249, dtype: float64

In [25]: avocado[["AveragePrice", "Date"]]

Out[25]:

	AveragePrice	Date
0	1.33	2015-12-27
1	1.35	2015-12-20
2	0.93	2015-12-13
3	1.08	2015-12-06
4	1.28	2015-11-29
...	...	...
18244	1.63	2018-02-04
18245	1.71	2018-01-28
18246	1.87	2018-01-21
18247	1.93	2018-01-14
18248	1.62	2018-01-07

18249 rows × 2 columns

In [26]: avocado[avocado["AveragePrice"]<1]

Out[26]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35
6	6	2015-11-15	0.99	83453.76	1368.92	73672.72	93.26	8318.86
7	7	2015-11-08	0.98	109428.33	703.75	101815.36	80.00	6829.22
13	13	2015-09-27	0.99	106803.39	1204.88	99409.21	154.84	6034.46
43	43	2015-03-01	0.99	55595.74	629.46	45633.34	181.49	9151.45
...	...	...	...	...	...	...	...	...
17169	43	2017-03-05	0.99	155011.12	35367.23	5175.81	5.91	114462.17
17170	44	2017-02-26	0.99	171145.00	34520.03	6936.39	0.00	129688.58
17536	39	2017-04-02	0.98	402676.23	34093.33	58330.53	207.85	310044.52
17537	40	2017-03-26	0.90	456645.91	36169.35	51398.72	139.55	368938.29
17540	43	2017-03-05	0.99	367519.17	61166.48	55123.99	126.80	251101.90

2796 rows × 14 columns



In [27]: avocado[avocado["type"]=="organic"]

Out[27]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Sr B
9126	0	2015-12-27	1.83	989.55	8.16	88.59	0.00	892.80	891
9127	1	2015-12-20	1.89	1163.03	30.24	172.14	0.00	960.65	960
9128	2	2015-12-13	1.85	995.96	10.44	178.70	0.00	806.82	806
9129	3	2015-12-06	1.84	1158.42	90.29	104.18	0.00	963.95	948
9130	4	2015-11-29	1.94	831.69	0.00	94.73	0.00	736.96	736
...	...	...	...	...	...	...	...	...	...
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13060
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9350
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10919
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11980

9123 rows × 14 columns



```
In [28]: avocado[(avocado["Date"]<"2015-02-04") & (avocado["type"]=="organic")]
```

Out[28]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
9173	47	2015-02-01	1.83	1228.51	33.12	99.36	0.0	1096.03	1096.03
9174	48	2015-01-25	1.89	1115.89	14.87	148.72	0.0	952.30	952.30
9175	49	2015-01-18	1.93	1118.47	8.02	178.78	0.0	931.67	931.67
9176	50	2015-01-11	1.77	1182.56	39.00	305.12	0.0	838.44	838.44
9177	51	2015-01-04	1.79	1373.95	57.42	153.88	0.0	1162.65	1162.65
...	...	...	...	...	...	...	...	...	...
11928	46	2015-02-01	1.77	7210.19	1634.42	3012.44	0.0	2563.33	2563.33
11929	47	2015-01-25	1.63	7324.06	1934.46	3032.72	0.0	2356.88	2320.00
11930	48	2015-01-18	1.71	5508.20	1793.64	2078.72	0.0	1635.84	1620.00
11931	49	2015-01-11	1.69	6861.73	1822.28	2377.54	0.0	2661.91	2656.66
11932	50	2015-01-04	1.64	6182.81	1561.30	2958.17	0.0	1663.34	1663.34

270 rows × 14 columns



```
In [29]: regionFilter = avocado["region"].isin(["Boston","SanDiego"])
         avocado[regionFilter]
```

Out[29]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags
208	0	2015-12-27	1.13	450816.39	3886.27	346964.70	13952.56	86012.86
209	1	2015-12-20	1.07	489802.88	4912.37	390100.99	5887.72	88901.80
210	2	2015-12-13	1.01	549945.76	4641.02	455362.38	219.40	89722.96
211	3	2015-12-06	1.02	488679.31	5126.32	407520.22	142.99	75889.78
212	4	2015-11-29	1.19	350559.81	3609.25	272719.08	105.86	74125.62
...	...	...	...	...	...	...	...	...
18100	7	2018-02-04	1.81	17454.74	1158.41	7388.27	0.00	8908.06
18101	8	2018-01-28	1.91	17579.47	1145.64	8284.41	0.00	8149.42
18102	9	2018-01-21	1.95	18676.37	1088.49	9282.37	0.00	8305.51
18103	10	2018-01-14	1.81	21770.02	3285.98	14338.52	0.00	4145.52
18104	11	2018-01-07	2.06	16746.82	5150.82	9366.31	0.00	2229.69

676 rows × 14 columns



```
In [30]: regionFilter = avocado['region'].isin(["Boston", "SanDiego"])
yearFilter = avocado["year"].isin(["2016", "2017"])
avocado[regionFilter & yearFilter]
```

Out[30]:

Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags
---------------	------	--------------	-----------------	------	------	------	---------------	---------------	---------------	----------------



```
In [33]: avocado.isna()
```

Out[33]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	X
0	False	False	False	False	False	False	False	False	False	False	
1	False	False	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	False	False	False	False	False	
3	False	False	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	False	False	
...	...	...	...	...	...	...	...	...	...	...	...
18244	False	False	False	False	False	False	False	False	False	False	
18245	False	False	False	False	False	False	False	False	False	False	
18246	False	False	False	False	False	False	False	False	False	False	
18247	False	False	False	False	False	False	False	False	False	False	
18248	False	False	False	False	False	False	False	False	False	False	

18249 rows × 14 columns



In [34]: avocado.isna().any()

```
Out[34]: Unnamed: 0      False
Date                False
AveragePrice        False
Total Volume        False
4046                 False
4225                 False
4770                 False
Total Bags          False
Small Bags          False
Large Bags          False
XLarge Bags         False
type                False
year                False
region              False
dtype: bool
```

In [35]: avocado.isna().sum()

```
Out[35]: Unnamed: 0      0
         Date          0
         AveragePrice  0
         Total Volume  0
         4046          0
         4225          0
         4770          0
         Total Bags    0
         Small Bags    0
         Large Bags    0
         XLarge Bags   0
         type          0
         year          0
         region        0
         dtype: int64
```

```
In [36]: avocado.dropna()
```

Out[36]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	6
...	...	...	...	...	...	...	...	...	...
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11

18249 rows × 14 columns




```
In [37]: meanVal = avocado["AveragePrice"].mean()  
avocado.fillna(meanVal)
```

Out[37]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	6
...	...	...	...	...	...	...	...	...	...
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11

18249 rows × 14 columns



```
In [38]: avocado["AveragePrice100"] = avocado["AveragePrice"] * 100  
avocado
```

Out[38]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	!
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	!
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	!
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	!
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	!
...	...	...	...	...	...	...	...	...	
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	1!
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	!
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	!
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	1!
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	1!

18249 rows × 15 columns



In [40]: avocado["AveragePrice"].mean()

Out[40]: 1.405978409775878

In [42]: avocado["Date"].max()

Out[42]: '2018-03-25'

```

In [73]: def pct30(column):
#return the 0.3 quartile
return column.quantile(0.3)
def pct50(column):
#return the 0.5 quartile
return column.quantile(0.5)

avocado[["AveragePrice", "Total Bags"]].agg([pct30, pct50])

```

Out[73]:

	AveragePrice	Total Bags
pct30	1.15	7316.634
pct50	1.37	39743.830

In [46]: `temp = avocado.drop_duplicates(subset=["year"])`
`temp`

Out[46]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87
2808	0	2016-12-25	1.52	73341.73	3202.39	58280.33	426.92	11432.09
5616	0	2017-12-31	1.47	113514.42	2622.70	101135.53	20.25	9735.94
8478	0	2018-03-25	1.57	149396.50	16361.69	109045.03	65.45	23924.33

In [47]: `avocado["year"].value_counts(sort=True, ascending = False)`

Out[47]:

year	count
2017	5722
2016	5616
2015	5615
2018	1296

Name: count, dtype: int64

In [74]: `# group by multiple columns and perform multiple summary statistic operations`
`avocado.groupby(["year", "type"])["AveragePrice"].agg([min, max, np.mean, np.median])`

C:\Users\ANIL\AppData\Local\Temp\ipykernel_30908\3377443975.py:2: FutureWarning: The provided callable <built-in function min> is currently using SeriesGroupBy.min. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "min" instead.

```
avocado.groupby(["year", "type"])["AveragePrice"].agg([min, max, np.mean, np.median])
```

C:\Users\ANIL\AppData\Local\Temp\ipykernel_30908\3377443975.py:2: FutureWarning: The provided callable <built-in function max> is currently using SeriesGroupBy.max. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "max" instead.

```
avocado.groupby(["year", "type"])["AveragePrice"].agg([min, max, np.mean, np.median])
```

C:\Users\ANIL\AppData\Local\Temp\ipykernel_30908\3377443975.py:2: FutureWarning: The provided callable <function mean at 0x00000157980C0C20> is currently using SeriesGroupBy.mean. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "mean" instead.

```
avocado.groupby(["year", "type"])["AveragePrice"].agg([min, max, np.mean, np.median])
```

C:\Users\ANIL\AppData\Local\Temp\ipykernel_30908\3377443975.py:2: FutureWarning: The provided callable <function median at 0x00000157981EB880> is currently using SeriesGroupBy.median. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "median" instead.

```
avocado.groupby(["year", "type"])["AveragePrice"].agg([min, max, np.mean, np.median])
```

Out[74]:

		min	max	mean	median
year	type				
2015	conventional	0.49	1.59	1.077963	1.08
	organic	0.81	2.79	1.673324	1.67
2016	conventional	0.51	2.20	1.105595	1.08
	organic	0.58	3.25	1.571684	1.53
2017	conventional	0.46	2.22	1.294888	1.30
	organic	0.44	3.17	1.735521	1.72
2018	conventional	0.56	1.74	1.127886	1.14
	organic	1.01	2.30	1.567176	1.55

In [75]: *# this is the same table we build in the previous cell but using pivot table*
 avocado.pivot_table(index=["year", "type"], aggfunc=[min, max, np.mean, np.median], val

C:\Users\ANIL\AppData\Local\Temp\ipykernel_30908\762502195.py:2: FutureWarning: The provided callable <built-in function min> is currently using DataFrameGroupBy.min. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "min" instead.

```
avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
```

C:\Users\ANIL\AppData\Local\Temp\ipykernel_30908\762502195.py:2: FutureWarning: The provided callable <built-in function max> is currently using DataFrameGroupBy.max. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "max" instead.

```
avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
```

C:\Users\ANIL\AppData\Local\Temp\ipykernel_30908\762502195.py:2: FutureWarning: The provided callable <function mean at 0x00000157980C0C20> is currently using DataFrameGroupBy.mean. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "mean" instead.

```
avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
```

C:\Users\ANIL\AppData\Local\Temp\ipykernel_30908\762502195.py:2: FutureWarning: The provided callable <function median at 0x00000157981EB880> is currently using DataFrameGroupBy.median. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "median" instead.

```
avocado.pivot_table(index=["year","type"], aggfunc=[min,max,np.mean,np.median], values="AveragePrice")
```

Out[75]:

		min	max	mean	median
		AveragePrice	AveragePrice	AveragePrice	AveragePrice
year	type				
2015	conventional	0.49	1.59	1.077963	1.08
	organic	0.81	2.79	1.673324	1.67
2016	conventional	0.51	2.20	1.105595	1.08
	organic	0.58	3.25	1.571684	1.53
2017	conventional	0.46	2.22	1.294888	1.30
	organic	0.44	3.17	1.735521	1.72
2018	conventional	0.56	1.74	1.127886	1.14
	organic	1.01	2.30	1.567176	1.55

In [51]: `regionIndex = avocado.set_index(["region"])`
`regionIndex`

Out[51]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770
region							
Albany	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16
Albany	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33
Albany	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50
Albany	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58
Albany	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78
...	...	...	...	...	...	...	...
WestTexNewMexico	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00
WestTexNewMexico	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00
WestTexNewMexico	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94
WestTexNewMexico	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01
WestTexNewMexico	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53

18249 rows × 14 columns



```
In [53]: avocado[avocado["region"].isin(["Albany", "WestTexNewMexico"])]
```

Out[53]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	6
...	...	...	...	...	...	...	...	...	...
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11

673 rows × 15 columns



```
In [54]: regionIndex.loc[["Albany", "WestTexNewMexico"]]
```

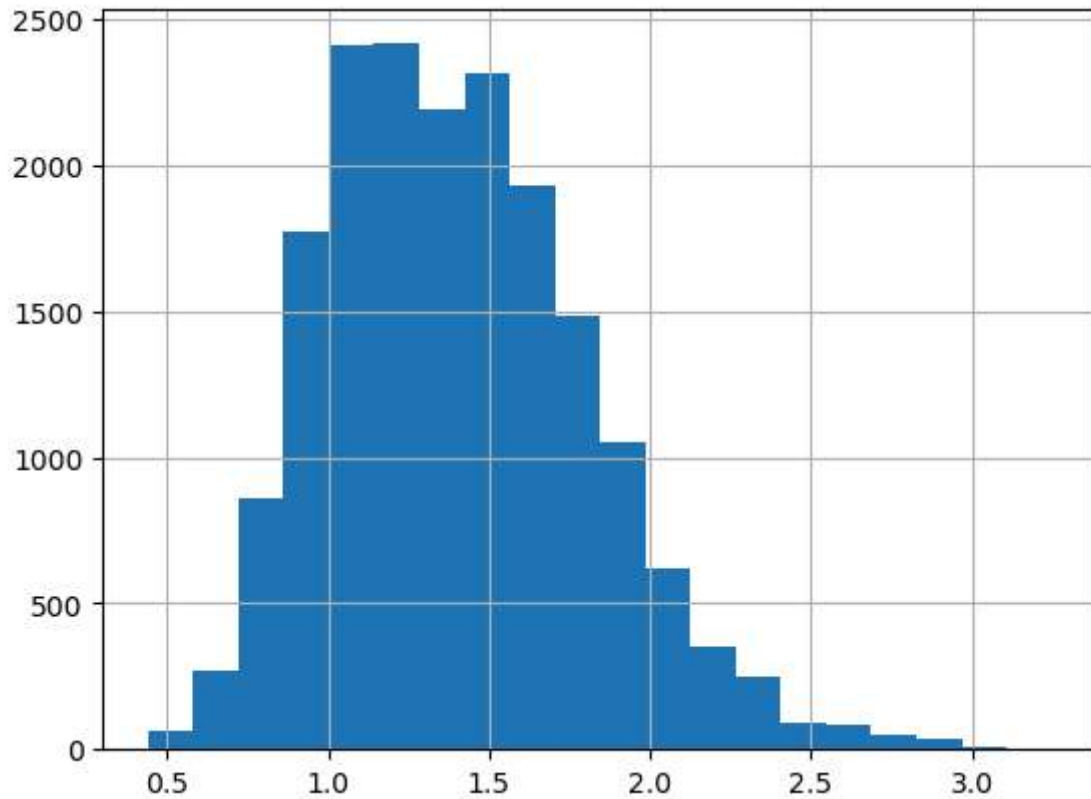
Out[54]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770
region							
Albany	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16
Albany	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33
Albany	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50
Albany	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58
Albany	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78
...	...	...	...	...	...	...	...
WestTexNewMexico	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00
WestTexNewMexico	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00
WestTexNewMexico	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94
WestTexNewMexico	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01
WestTexNewMexico	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53

673 rows × 14 columns



```
In [55]: avocado["AveragePrice"].hist(bins=20)
plt.show()
```

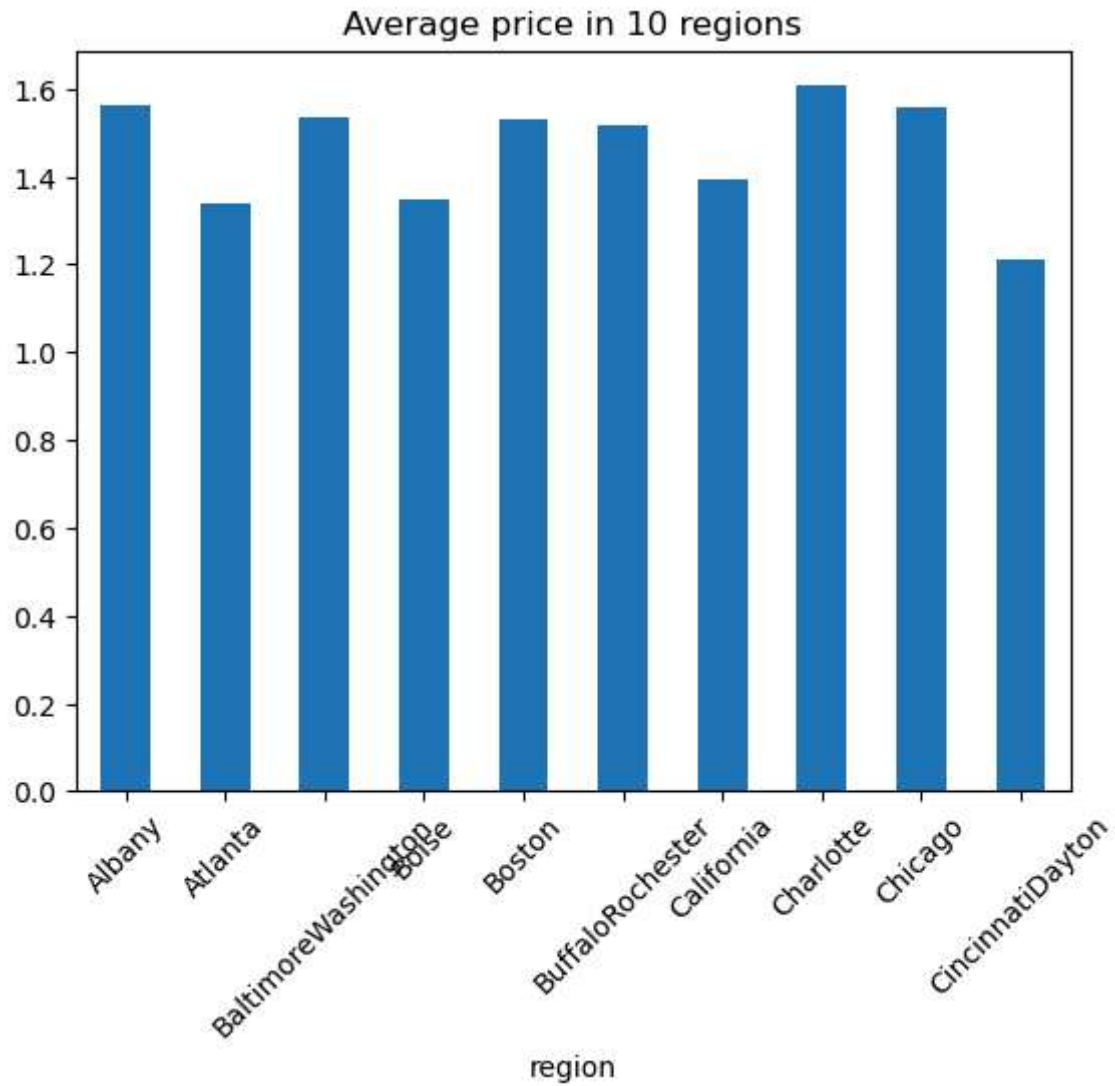
```
In [58]: regionFilter = avocado.groupby("region")["AveragePrice"].mean().head(10)
regionFilter
```

```
Out[58]: region
Albany          1.561036
Atlanta         1.337959
BaltimoreWashington  1.534231
Boise           1.348136
Boston          1.530888
BuffaloRochester  1.516834
California      1.395325
Charlotte       1.606036
Chicago         1.556775
CincinnatiDayton  1.209201
Name: AveragePrice, dtype: float64
```

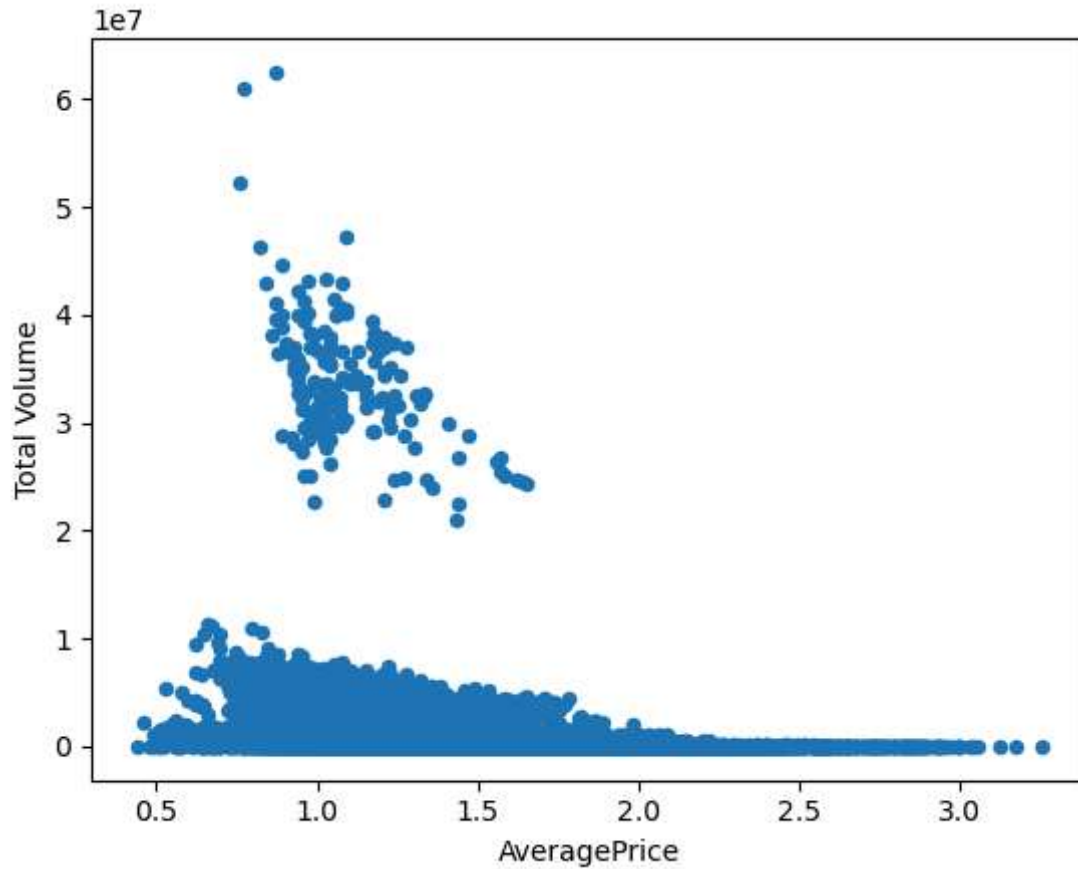
```
In [61]: #regionFilter.plot(kind = "bar",rot=45,title="Average price in 10 regions")
```

```
Out[61]: <Axes: title={'center': 'Average price in 10 regions'}, xlabel='region'>
```

```
In [63]: regionFilter.plot(kind = "bar",rot=45,title="Average price in 10 regions")
plt.show()
```



```
In [66]: avocado.plot(x="AveragePrice", y="Total Volume", kind="scatter")  
plt.show()
```



```
In [67]: avocado["AveragePrice"].sub(avocado["AveragePrice"])
```

```
Out[67]: 0      0.0
         1      0.0
         2      0.0
         3      0.0
         4      0.0
         ...
        18244    0.0
        18245    0.0
        18246    0.0
        18247    0.0
        18248    0.0
        Name: AveragePrice, Length: 18249, dtype: float64
```

```
In [ ]:
```